

IA: MACHINE LEARNING FOUNDATIONS

SESSION 3 – Types of Learning

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AGENDA

1. Frequent Learning Methods
2. Other types of learning
3. Regression vs Classification
4. Perceptron Learning Algorithm



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1. FREQUENT LEARNING METHODS

1.1. SUPERVISED LEARNING

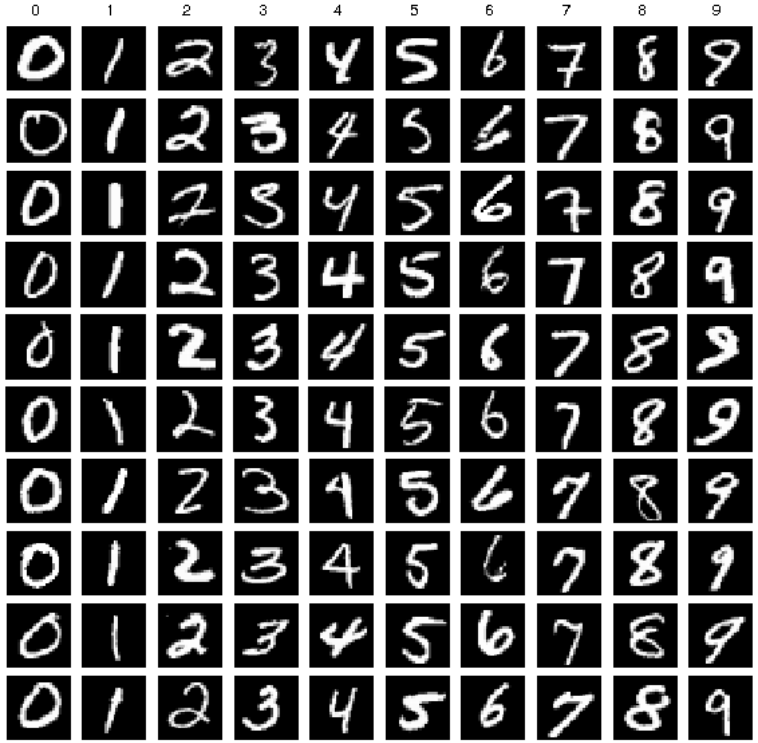
- When the training data contains explicit examples of what the correct output should be for given inputs.
- All the examples seen so far.

TRAINING EXAMPLES
$(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)$

data set $\mathcal{D}$
where $y_n = f(\mathbf{x}_n)$ for $n = 1, \ldots, N$

- In many cases, manual labeling is needed, and it is very time consuming.
- Most popular in real problems

(image , digit)



$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

1. FREQUENT LEARNING METHODS

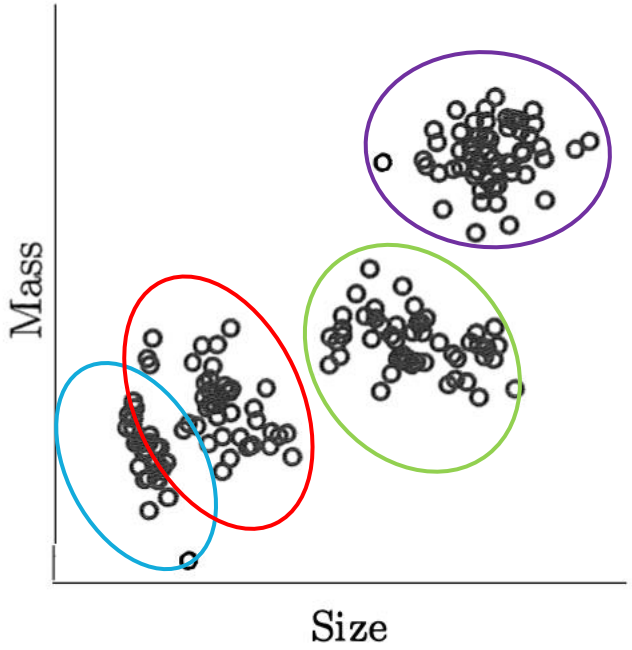
1.2. UNSUPERVISED LEARNING

- The training data does not contain any output information at all: *unlabeled data*.

$$\mathbf{x}_1, \cdots, \mathbf{x}_N$$

- The idea is to find relationships between data by leveraging their features.

- It can also be seen as a way to create a higher-level representation of the data. Indeed, unsupervised learning can be viewed as a precursor of supervised learning. Two examples:
  - Learning a language
  - Autoencoders



1. FREQUENT LEARNING METHODS

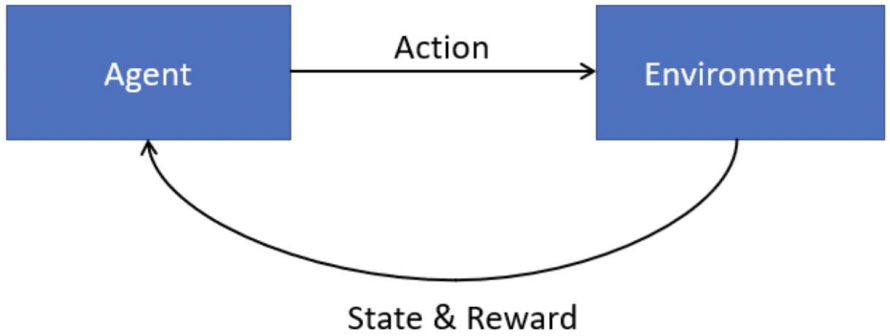
1.3. REINFORCEMENT LEARNING



"Google's AlphaZero went from only knowing the rules of chess to beating the best players in the world just in 9 hours"



- Reinforcement learning models learn from an **environment**.
- The **environment** has a set of rules and is usually assumed to be deterministic.
- A reinforcement learning model interacts with the **environment** through an **agent**.
- The **agent** has a **state** in the **environment**, and **the agent can perform actions that change the state of the agent in the environment**.



1. FREQUENT LEARNING METHODS

1.3. REINFORCEMENT LEARNING

CHESS EXAMPLE

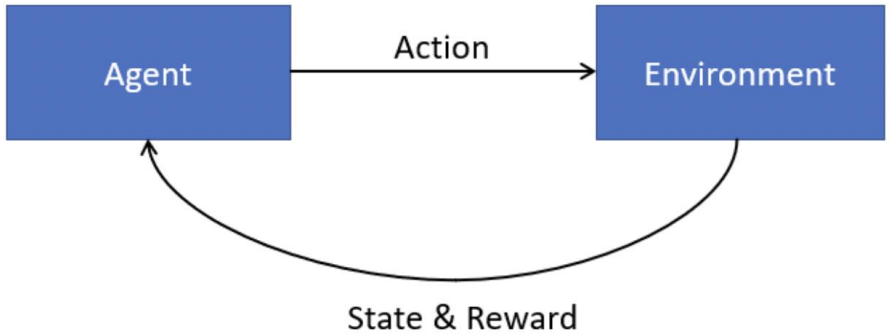
- The **environment** is the chessboard.
- The **agent** is the chess bot.
- The **state** of the environment is the position of all the pieces.

- Given the state of the chessboard, there are only a finite number of legal moves (**actions**) that can be made, these are determined by the environment. For example, the king can't move the same way as the queen.

- When the agent takes an action, the environment will receive this as an input and will output the resulting **state and a reward**.

- After the agent moves a piece, the environment would return what the chessboard looks like after this move and the opponent's move, this way it's the agent's turn again. The environment will also return a reward, say if you capture a piece for example.





EXERCISE

For each of the following tasks, identify which type of learning is involved:

a) Recommending a book to a user in an online bookstore.
b) Playing backgammon.
c) Grouping movies into different types.
d) Learning to play music.
e) Credit limit: Deciding the maximum allowed debt for each bank customer.

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SOLUTION

(a) Supervised Learning
(b) Reinforcement Learning
(c) Unsupervised Learning
(d) Learning to play music

If learn by yourself, it's unsupervised learning
If learn from a teacher, it's supervised learning
If learn by yourself but with someone to tell you if your music is good or not, it's reinforcement learning.

(e) Supervised Learning

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4. OTHER TYPES OF LEARNING

- In recent years, many learning techniques have emerged in response to the big number of real problems:

Semisupervised Learning

It combines a small amount of labeled data with a large amount of unlabeled data during training

Multitask Learning

Technique in which we try to learn multiple tasks simultaneously, optimizing multiple loss functions at once.

Online Learning

Data becomes available in a sequential order and is used to update the best predictor for future data at each step

Transfer Learning

It focuses on storing knowledge gained while solving one problem and applying it to a different but related problem

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5. REGRESSION VS CLASSIFICATION

Quantitative variables **vs** **Qualitative variables**



5. REGRESSION VS CLASSIFICATION

Quantitative variables

They take on numerical values:

- Person's age, height or income
- Value of a house
- Price of a stock

REGRESSION

vs

Qualitative variables

They take values in one of *K classes* or categories:

- Person's marital status (marriaged or not)
- Brand of producto (A, B or C)
- Cancer diagnosis (healthy or not)

CLASSIFICATION

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CLASSIFICATION

- Most of the machinel learning methods discussed in this course can be applied regardless of the predictor variable type.

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6. PERCEPTRON LEARNING ALGORITHM

$\mathcal{X} = \mathbb{R}^d$

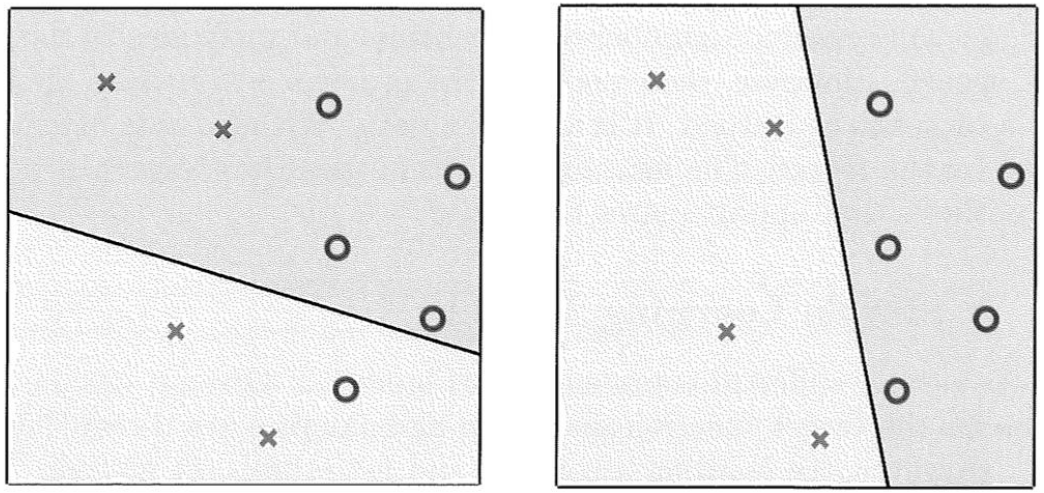
$\mathcal{Y} = \{+1, -1\}$

Approve credit if $\sum_{i=1}^{d} w_i x_i > \text{threshold}$

Deny credit if $\sum_{i=1}^{d} w_i x_i < \text{threshold}$

$$h(\mathbf{x}) = \text{sign}\left(\left(\sum_{i=1}^{d} w_i x_i\right) + b\right) \longrightarrow h(\mathbf{x}) = \text{sign}(\mathbf{w}^T\mathbf{x})$$

$w_1, \cdots, w_d$



(a) Misclassified data

(b) Perfectly classified data

The figure illustrates what a perceptron does in a two-dimensional case (d=2).

The plane is Split by a line into two regions (+1 and -1)

Different values for the parameters w_1, w_2, b Correspond to different lines:

$$w_1 x_1 + w_2 x_2 + b = 0$$

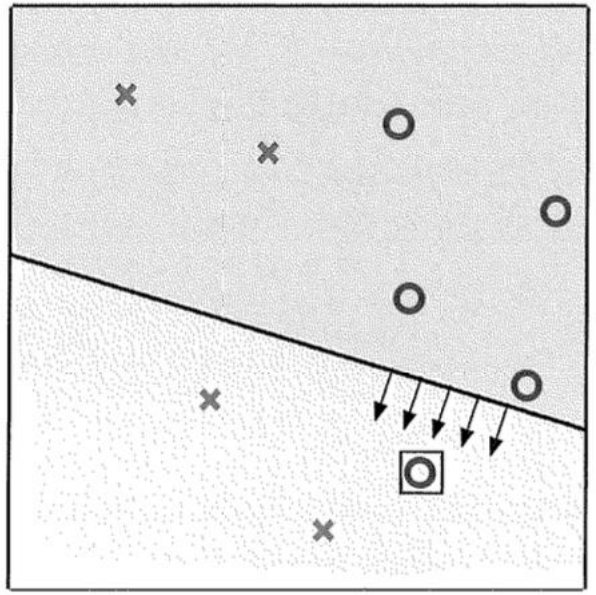
6. PERCEPTRON LEARNING ALGORITHM

- The goal is to find the optimal weights ($\mathbf{w}$) that find the best threshold.
- The algorithm will find this weights ($\mathbf{w}$) using a simple iterative method:
  - At iteration t we have $\mathbf{w}(t)$
  - The algorithm picks an example from $(\mathbf{x}_1, y_1) \cdots (\mathbf{x}_N, y_N)$ that is currently misclassified: $(\mathbf{x}(t), y(t))$
  - The weights are updated following:

$$\mathbf{w}(t+1) = \mathbf{w}(t) + y(t)\mathbf{x}(t)$$

$$y(t) \neq \text{sign}(\mathbf{w}^{\text{T}}(t)\mathbf{x}(t))$$

  - The algorithm continues with further iterations until there are no misclassified samples.



6. PERCEPTRON LEARNING ALGORITHM

PROBLEM

Create a target function f and a bidimensional dataset D and see how the perceptron learning algorithm works. Choose a random line in the plane as your target function, where one side of the line maps to +1 and the other maps to -1. Choose the inputs x_n of the data set as random points in the plane and evaluate the target function on each x_n to get the corresponding output y_n.

a) Create a dataset of size 100. Run the perceptron learning algorithm on your data set and see how long it takes to converge and how well the final hypothesis g matches your target f.
b) Plot the examples $\{(x_n, y_n)\}$ as well as the target function f on a plane. Be sure to mark the examples from different classes differently.
c) Report the number of updates that the algorithm takes before converging.
d) Repeat everything in (a) with another randomly generated data set of size 1000. Compare the results with (a).
e) Modify the algorithm such that it takes $x_n \in \mathbb{R}^{10}$. Randomly generate a linearly separable data set of size 1000 with $x_n \in \mathbb{R}^{10}$ and feed the data set to the algorithm. How many updates does the algorithm take to converge?


THANK YOU